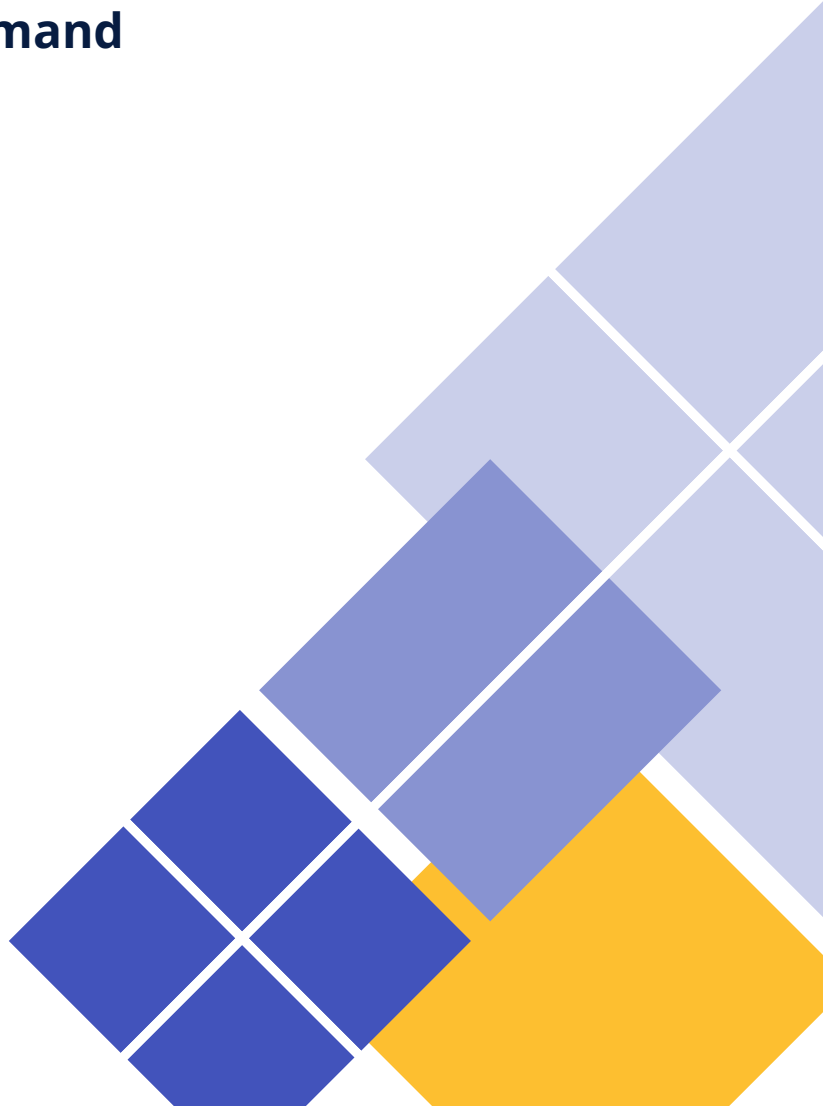




MONITORING AND RETRAINING DOCUMENTATION

**Sales Forecasting and Demand
Prediction Project**

2025



INTRODUCTION

In real-world production environments, machine learning models do not remain accurate forever.

Data changes, customer behavior evolves, seasons shift, and external factors such as weather and economic conditions play a major role in shaping demand.

This documentation explains two essential components that ensure the forecasting system continues to perform reliably over time:

- **Model Monitoring**
- **Model Retraining Strategy**

Together, they ensure long-term accuracy, stability, and business value after deployment.

Model Monitoring

CONCEPT OVERVIEW

Model Monitoring is the continuous process of tracking the performance, stability, and data quality of a deployed model.

Its goal is to ensure the model remains accurate and aligned with real-world conditions.

In simple terms:

Model monitoring continuously checks whether the model is still producing accurate and reliable predictions.

Problems Monitoring Solves

Performance Degradation

Model accuracy declines over time due to new patterns not seen in training.

1

Data Drift

The statistical distribution of input data changes.

This affects how well the model understands new data.

2

Concept Drift

The relationship between features and the target changes.

This occurs when customer behavior or market dynamics evolve.

3

Monitoring Workflow

1. Collect New Incoming Data
2. Model Generates Predictions
3. Store Predictions
4. Real Sales Values Arrive
5. Calculate Performance Metrics
6. Detect Issues missing data, or anomalies.
7. Trigger Alerts

Alerting System

Alerts are classified into:

- **Normal** → System stable
- **Warning** → Performance slightly declining
- **Critical** → Drift detected or accuracy below threshold

Alerts notify:

- Data Science team
- Business stakeholders
- IT/DevOps team

Model Retraining Strategy

CONCEPT OVERVIEW

Model retraining is the process of updating the model using new, more recent data.

Its purpose is to keep the model aligned with current customer behavior and business conditions.

Why Do We Retrain

Retraining is necessary when

1. Data drift occurs
2. Concept drift appears
3. Model accuracy declines
4. New products, cities, or categories are introduced
5. External factors significantly change

When Should We Retrain

A. Scheduled Retraining

Performed on a fixed schedule:

- Weekly
- Monthly
- Quarterly

B. Trigger-Based Retraining

Automatically triggered when:

- Accuracy drops below threshold
- Drift is detected
- Significant business changes occur

C. Data-Volume-Based Retraining

Retrain after enough recent data has been collected (e.g., last 90 days).



Types of Retraining

Retraining is necessary when

1. Full Retraining

Train the model from scratch using all available historical data.
Used when major drift or market changes occur.

2. Incremental Retraining

Train using only new data.
Used when updates must be frequent and lightweight.



THANK YOU

